

EXPERIMENT 03

Password Capturing Using Wireshark

OBJECTIVE

To capture and analyze network traffic using Wireshark network protocol analyzer, inspect communication protocols, evaluate packet headers across network layers, and analyze data streams for credential exposure and encryption posture.

TOOLS / REQUIREMENTS

- **Forensic Software:** Wireshark Network Protocol Analyzer (Versions 4.6.4 and 4.6.8)
- **Operating Environments:** Linux Virtual Machine / Microsoft Windows 11
- **Network Interface:** Wireless Network Adapter (Wi-Fi / `enp0s3`)

EXPERIMENT SCENARIO

Controlled laboratory evidence was used for this experiment.

A packet sniffing and protocol analysis scenario was executed to evaluate network traffic on active interfaces. The investigator captured live packets, applied display filters (`dns`, `ip`, `tcp`, `udp`), analyzed deep frame headers across OSI layers, followed TCP streams, and examined the protocol distribution hierarchy.

EVIDENCE / INPUT

- **Capture Interface:** Wi-Fi (`\\Device\\NPF_{37BBBAAB-CB2A-4351-9D61-0CA28A47E4ED}`)
- **Local Host IP / MAC:** `192.168.1.37 / 58:02:05:4c:2f:b8`
- **Total Packets Captured:** 755 packets (536 TCP/TLS packets analyzed)
- **Target Protocols:** Ethernet II, IPv4, IPv6, TCP, UDP, DNS, TLSv1.2, SSDP

PROCEDURE

Step 1 — Wireshark Interface Selection in Linux Virtual Machine

Wireshark 4.6.4 was launched in a Linux virtual environment to examine available capture interfaces, highlighting network adapter `enp0s3`.

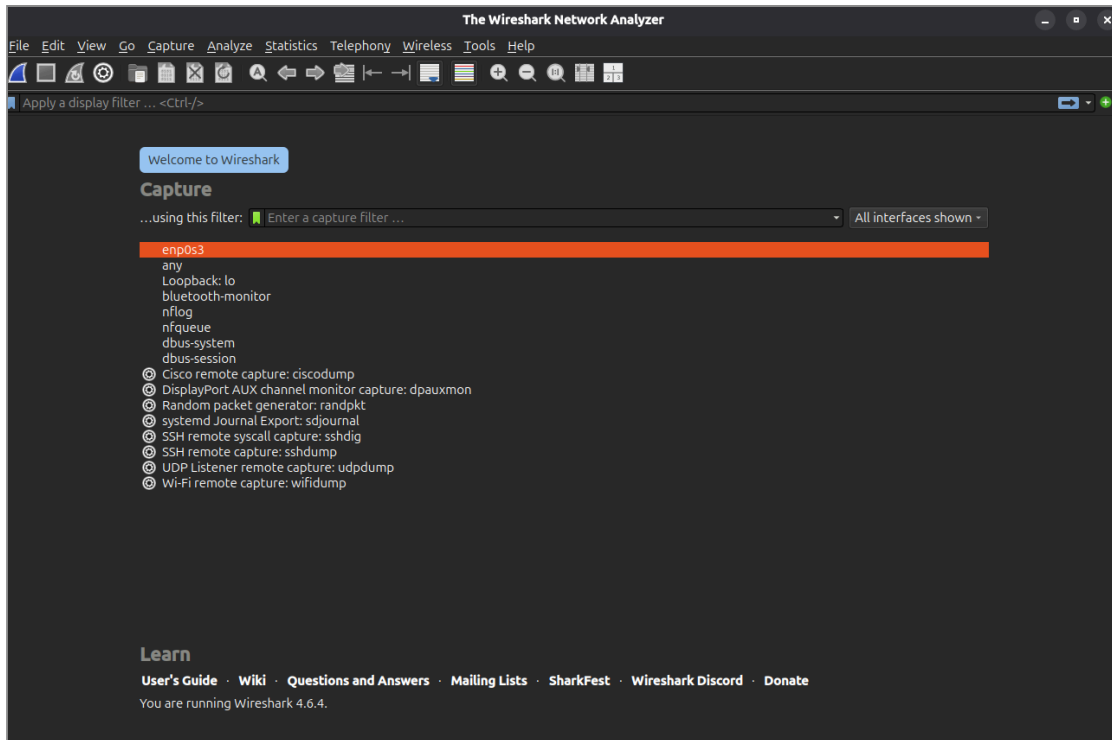


Figure 3.1: Wireshark capture interface selection in Linux VM

Step 2 — Wireshark Initialization on Windows Host

Wireshark 4.6.8 was opened on the Windows host, displaying active interface traffic activity graphs on the Wi-Fi adapter (IP 169.254.93.188 / `fe80::b38c:733c:d43:1aea`).

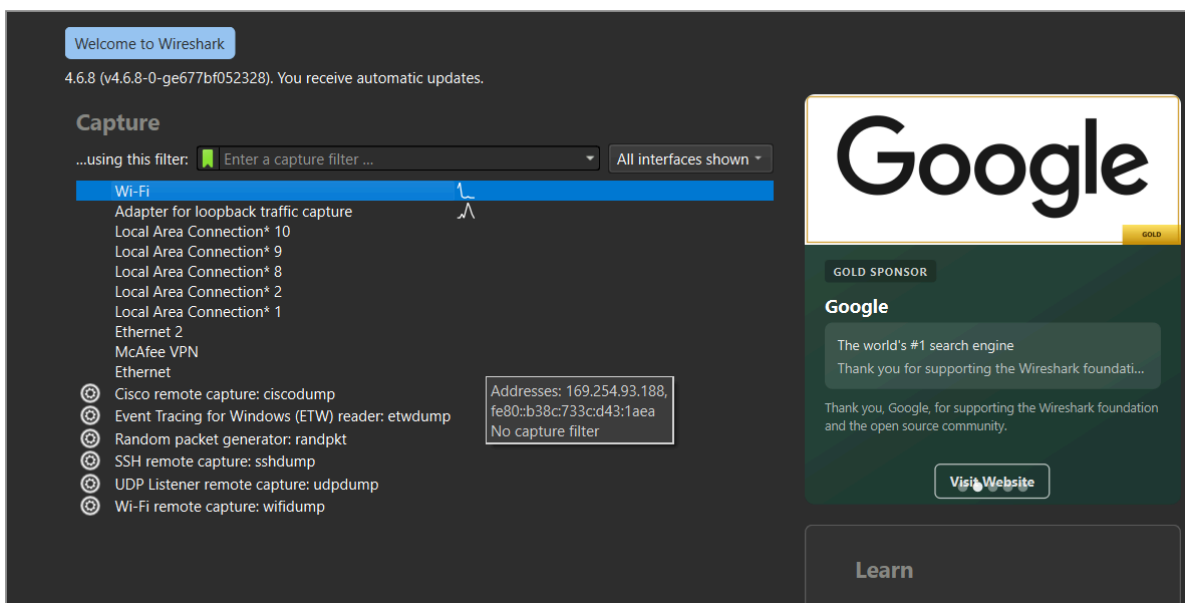


Figure 3.2: Wireshark welcome screen and Wi-Fi interface selection on Windows

Step 3 — Initiating Live Packet Capture on Wi-Fi Interface

Packet capturing was initiated on the Wi-Fi interface. Live network traffic involving local host 192.168.1.37 and external nodes was captured in real time.

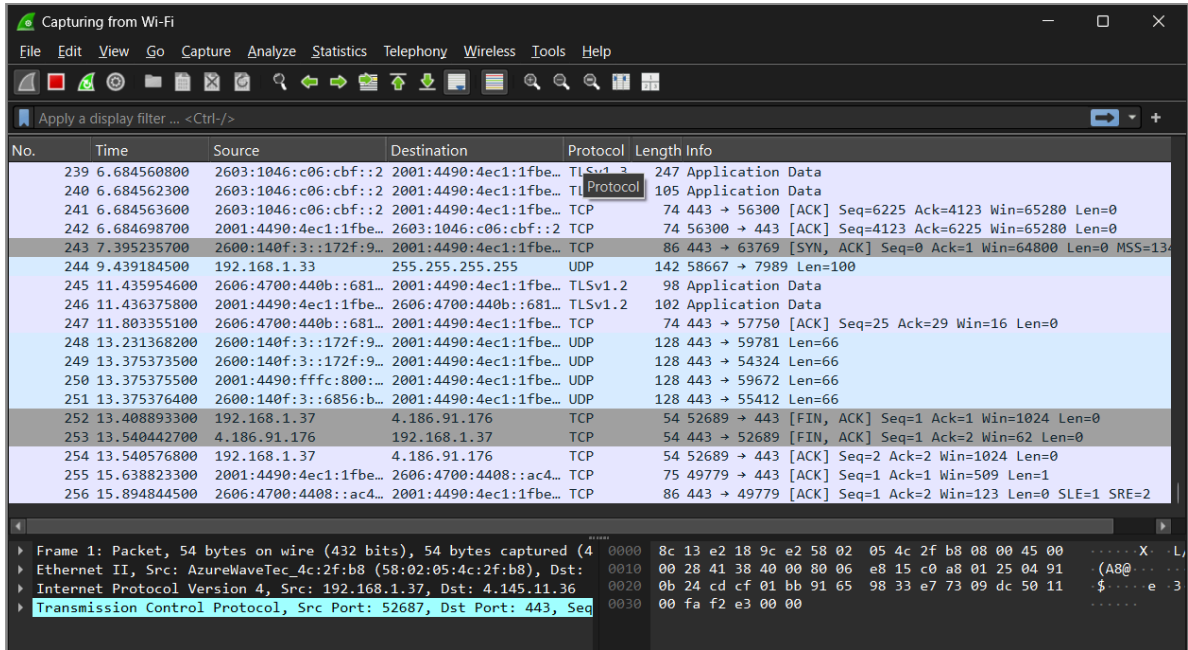


Figure 3.3: Live packet capture on Wi-Fi interface

Step 4 — Overview of Captured Transmission Frames

Captured packets were reviewed, showing standard TCP three-way handshake segments, TLSv1.2 Server Hello, Client Key Exchange, and connection termination flags (FIN/ACK, RST/ACK).

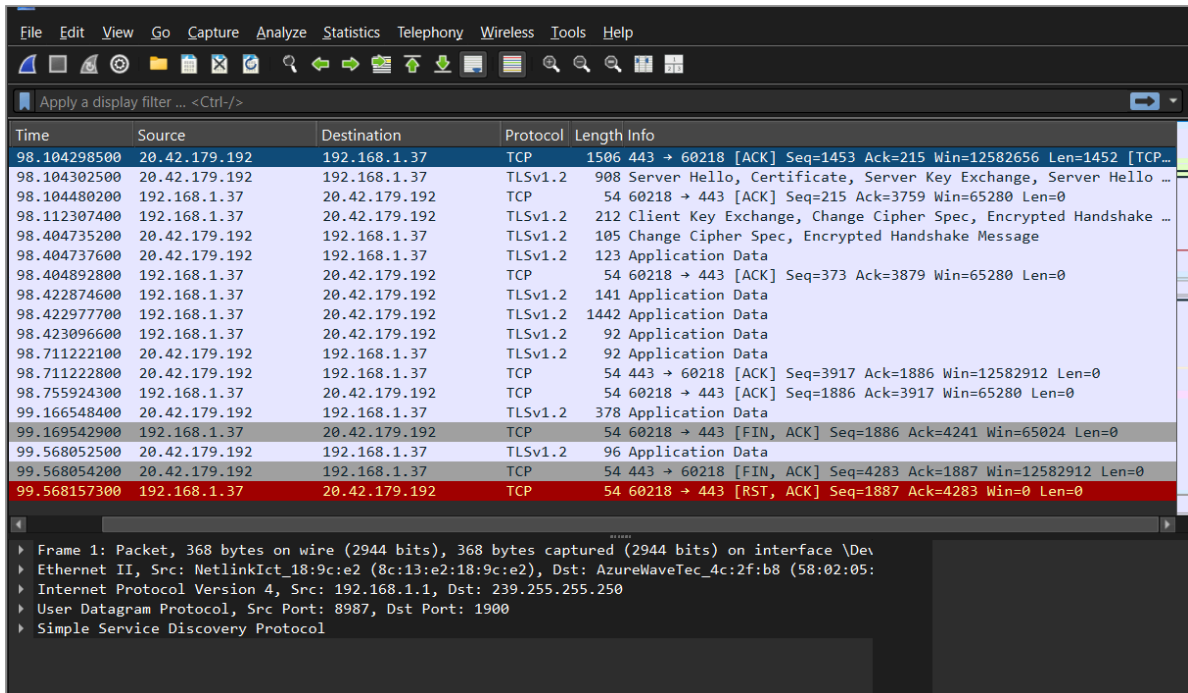


Figure 3.4: Overview of captured transmission frames

Step 5 — Filtering DNS Protocol Queries and Responses

The display filter `dns` was applied to isolate domain name resolution traffic, identifying standard queries and A/AAAA responses for `google.com`, `chatgpt.com`, and `github.com`.

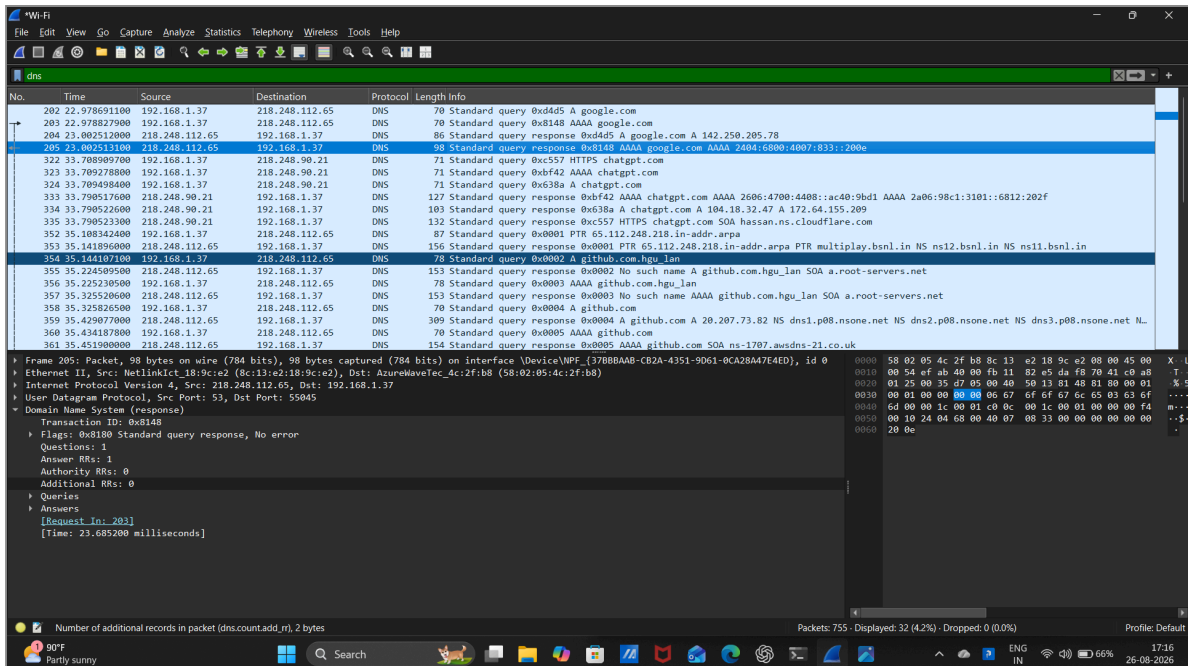


Figure 3.5: Applied DNS display filter showing domain queries

Step 6 — Filtering Internet Protocol (IPv4/IPv6) Traffic

The display filter `ip` was applied to filter network-layer packets between local gateway `192.168.1.1`, host `192.168.1.37`, and DNS resolvers (`218.248.112.65`).

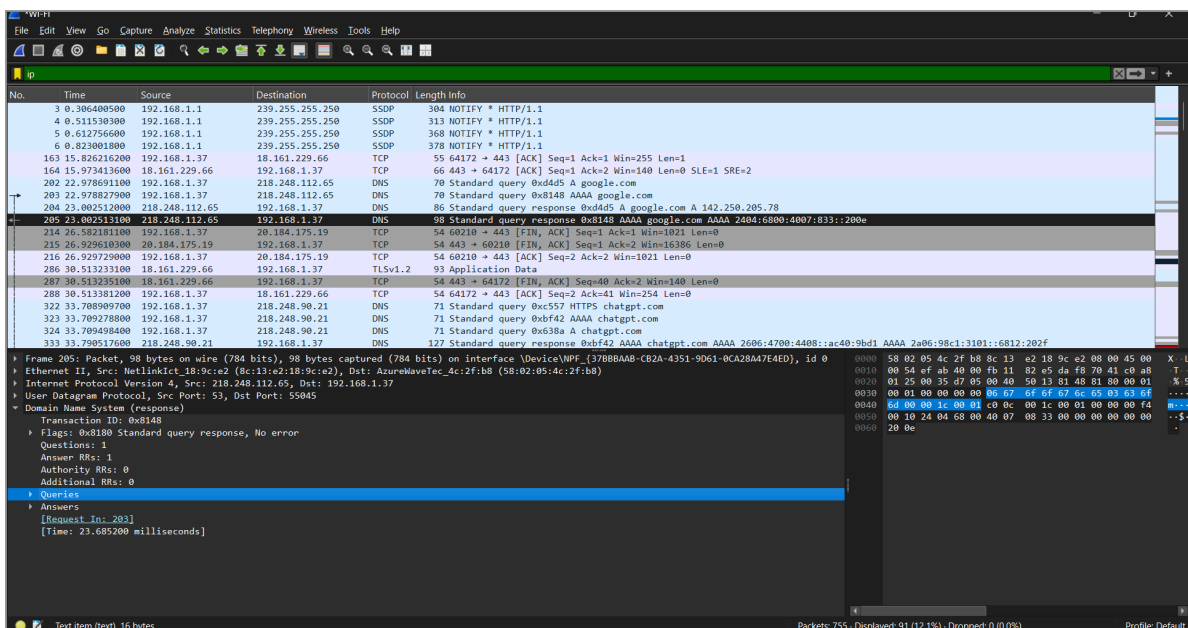


Figure 3.6: Applied IP display filter

Step 7 — Filtering Transmission Control Protocol (TCP) Streams

The display filter `tcp` was applied to analyze transport-layer communication streams, TCP sequence/acknowledgment numbers, and reassembled TCP PDUs.

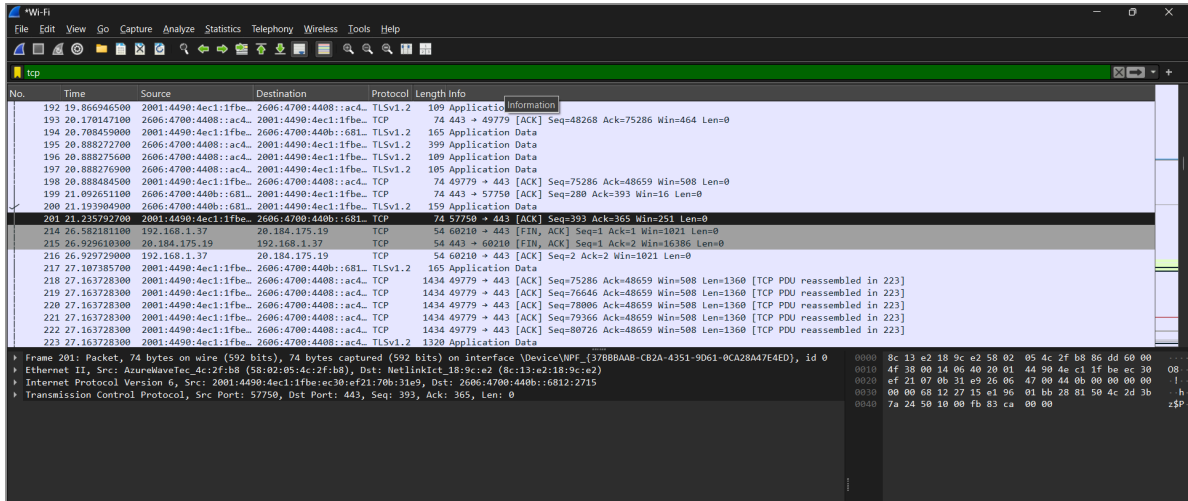


Figure 3.7: Applied TCP display filter

Step 8 — Filtering User Datagram Protocol (UDP) Traffic

The display filter `udp` was applied to inspect connectionless datagram traffic, revealing Simple Service Discovery Protocol (SSDP) multicast notifications on destination `239.255.255.250:1900`.

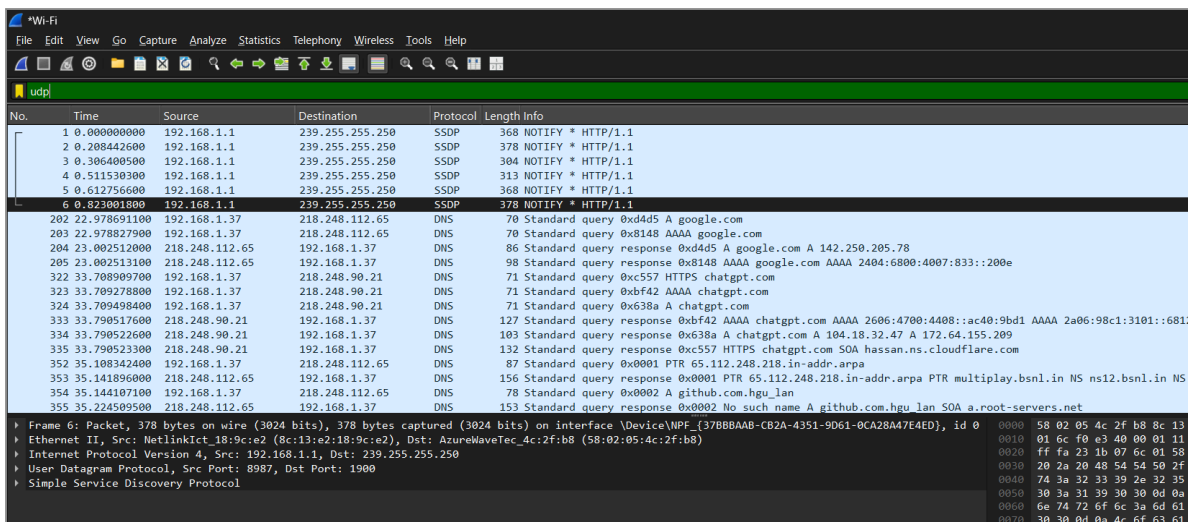


Figure 3.8: Applied UDP display filter

Step 9 — Deep Packet Inspection of Transport Layer Security Frame

Packet 14 was selected to examine detailed header fields: Frame (1132 bytes), Ethernet II, IPv6 (2001:4490:4ec1:1fb:ec30:ef21:70b:31e9, Dst: 2606:4700:4408::ac40:9bd1 under TLS.

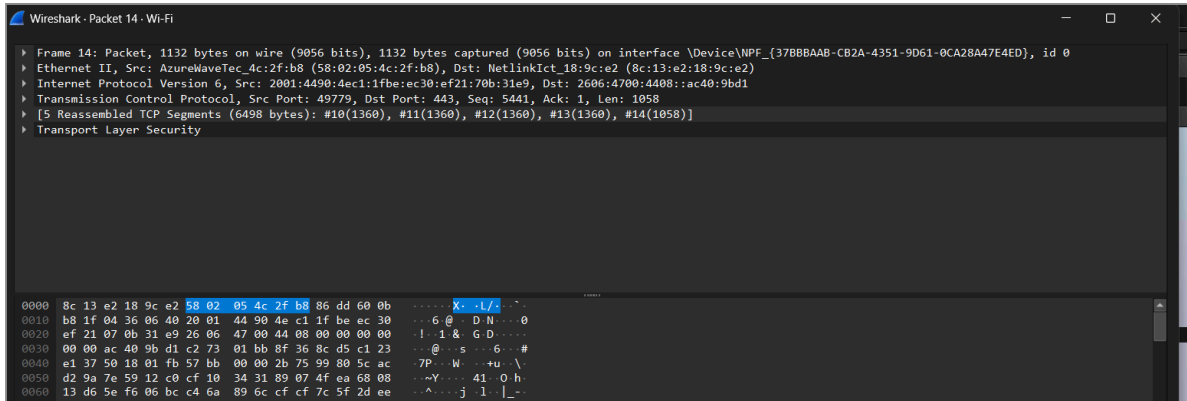


Figure 3.9: Deep packet inspection of TLS frame 14

Step 10 — Following TCP Stream Application Payload

The *Follow TCP Stream* feature (tcp.stream eq 1) was executed to inspect the raw data stream. The stream payload confirmed active cryptographic encryption preventing plaintext credential extraction.

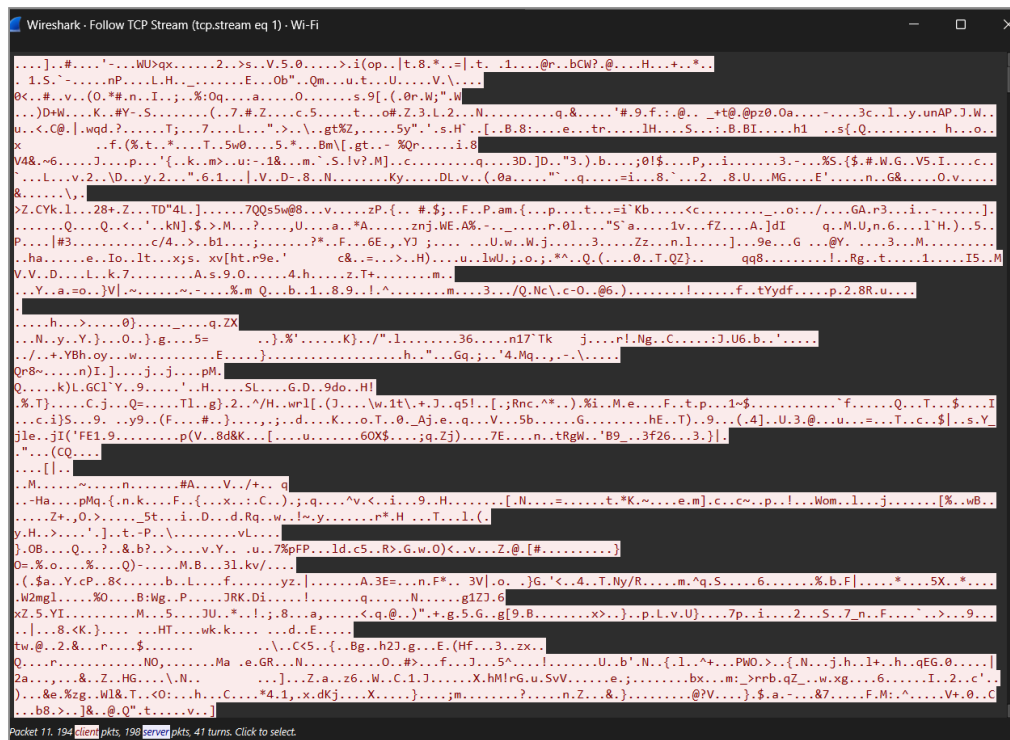
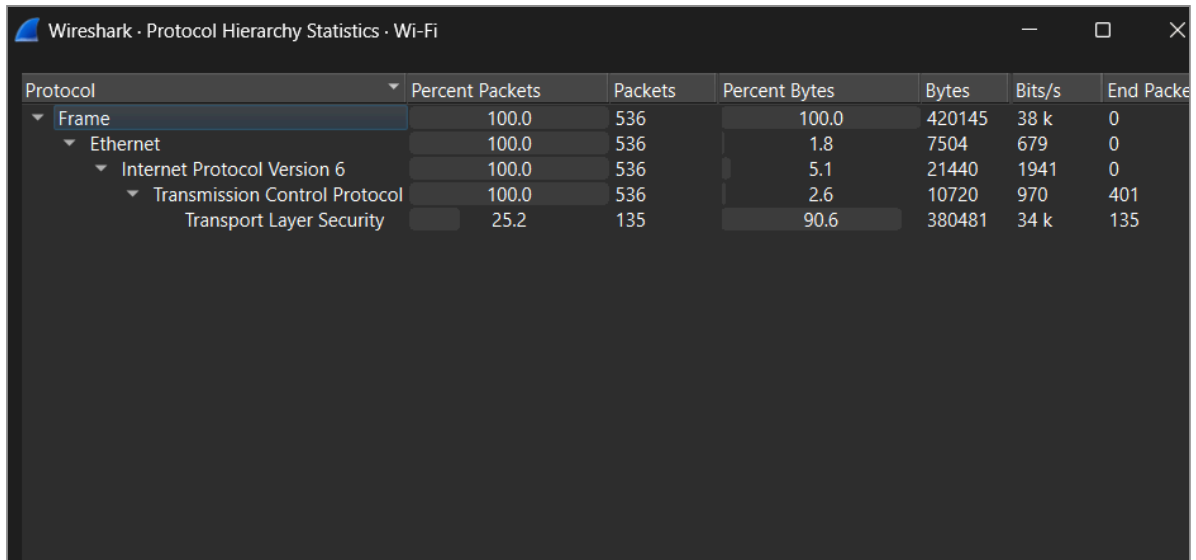


Figure 3.10: Follow TCP Stream payload showing encrypted session data

Step 11 — Protocol Hierarchy Statistics Analysis

The *Protocol Hierarchy Statistics* window was opened, detailing packet distribution: 100% Ethernet/IPv6/TCP traffic, with Transport Layer Security (TLS) accounting for 90.6% of transmitted bytes.



The image shows the 'Wireshark - Protocol Hierarchy Statistics - Wi-Fi' window. It displays a tree view of protocols and their corresponding statistics. The data is as follows:

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packet
Frame	100.0	536	100.0	420145	38 k	0
Ethernet	100.0	536	1.8	7504	679	0
Internet Protocol Version 6	100.0	536	5.1	21440	1941	0
Transmission Control Protocol	100.0	536	2.6	10720	970	401
Transport Layer Security	25.2	135	90.6	380481	34 k	135

Figure 3.11: Protocol hierarchy statistics distribution window

OBSERVATIONS

1. 755 total packets were captured across the active Wi-Fi interface.
2. DNS resolution traffic tracked standard queries and responses for `google.com`, `chatgpt.com`, and `github.com`.
3. Transport Layer Security (TLS) represented 90.6% of network traffic payload.
4. Following TCP Stream displayed encrypted cryptographic ciphertexts rather than plaintext form inputs.

FINDINGS

- **Technical Protocol Evaluation:** The laboratory exercise evaluated packet traffic across network layers. In the empirical capture session, modern web traffic utilized HTTPS/TLS encryption over port 443. Consequently, data payloads were encrypted, preventing plaintext credential interception and demonstrating the forensic impact of transport layer encryption.
- **Protocol Analysis Capability:** Wireshark effectively dissected traffic across OSI layers 2 through 7 (Ethernet II -> IPv4/IPv6 -> TCP/UDP -> TLS/DNS).

RUBRICS

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

RESULT

Successfully Completed. Wireshark successfully captured live network packets, filtered protocols across layers 3–7, analyzed stream payloads, and generated protocol hierarchy statistics.

CONCLUSION

Wireshark captured and analyzed live network traffic across multiple protocols. Inspection of the TCP streams and protocol hierarchy statistics demonstrated that active TLS encryption protected application payloads from plaintext credential sniffing.

REFERENCES

- Wireshark User's Guide (Wireshark Foundation)
- Digital Forensics Laboratory Manual: Password Capturing